$$B = \frac{1}{N} \sum_{i=1}^{N} (f_i - o_i)^2 \quad (3)$$

where N is the test set size, f_i is the forecasted probability of each event i, and $o_i \in \{0, 1\}$ its outcome.

Calibration plots can be seen in Figure 3. Each bubble represents a subset of forecasts, grouped by estimated probability; the number on each bubble is the size of the subset. Our plot reports shades of colour, where light means low uncertainty and dark means high uncertainty. The correlation between the probability returned by the models and the forecast conversion rate, i.e. the actual rate of forecasts turning true, can be seen in both charts, although it is higher in ours.

TABLE I
EVALUATION RESULTS FOR THE TWO APPROACHES. LOWER IS BETTER. OTHER BRIER SCORES ARE GIVEN FOR REFERENCE.

Reference	**Brier score**
Humans	0.200–0.300
Prediction markets	0.100–0.200
Manifold (480 days)	0.182
Average superforecasters	0.150
Evaluation results	**Brier score**
Random chance	0.250
Vanilla GPT-4o	0.236
Probability Estimator (ours)	0.186

Table I shows the evaluation results for our Probability Estimator against vanilla GPT-4o. We achieve a Brier score of 0.186, which means a +19% improvement over the widely-available AI and a +26% improvement over random chance. These results confirm that the knowledge condensed in the trends' trajectories as well as the ability of estimating uncertainties do provide strategic advantage.

The table also features additional reference ranges from Calibration City[1]. According to the evaluation, our Probability Estimator could score better than some prediction markets and would be only 0.036 Brier points away from the average superforecaster. Moreover, Calibration City reports forecasts whose market open length is between 0 and 480 days and uses the probability at market midpoint as the value to test. As the time between forecasting and fact-checking in our dataset is about 240 days, we would need to compare with markets twice as long. The only platform having an analogous magnitude of data ($N = 39$) for 480-day-long markets is Manifold. With a Brier score of 0.182, this result is in line with the score obtained by our Probability Estimator on our own dataset. Although these last results are not directly comparable, we believe they are significant and strongly argue in favour of autonomous LLM-based forecasting.

In the future, even the semantic dimension may be used to calibrate probabilities better. Figure 4 shows that for topics such as *Climate Change*, forecast probabilities were underestimated, as opposed to those in topics such as *Energy* which were overestimated.

[1]https://calibration.city/

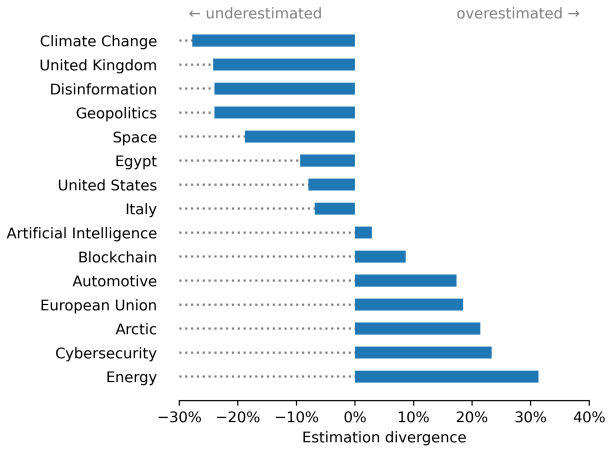

Fig. 4. Probability estimation performance by topic.

V. CONCLUSION

In this paper, we have introduced a novel approach to AI-driven foresight based on current trends, LLMs, and their logarithmic probabilities. Despite recent controversies regarding the ability of language models to perform forecasts accurately, we believe there is now enough evidence for this possibility. We found our systems are not yet performing at the same level as superforecasters. However, our results confirm that the knowledge condensed in trends and uncertainty values do provide strategic advantage to the end user.

This work may enable several applications in strategic foresight and adjacent fields. After testing the quality of the returned probabilities, future evaluations may target the utility of the forecast themselves. By exploiting the visionary capability of our Forecast Generator, we can create complex multi-forecast scenarios and analyse their interactions with the environment. Simulations by LLM-powered agents can become more reliable having a system which can compute probabilities of virtually any kind of event. Finally, new avenues may be opened for data-driven decision making in a personal, enterprise, or public setting.

REFERENCES

[1] Halawi, D., Zhang, F., Yueh-Han, C. and Steinhardt, J., 2024. Approaching Human-Level Forecasting with Language Models. arXiv preprint arXiv:2402.18563.

[2] Hsieh, E., Fu, P. and Chen, J., 2024. Reasoning and Tools for Human-Level Forecasting. arXiv preprint arXiv:2408.12036.

[3] Hurst, A., Lerer, A., Goucher, A.P., Perelman, A., Ramesh, A., Clark, A., Ostrow, A.J., Welihinda, A., Hayes, A., Radford, A. and Madry, A., 2024. GPT-4o System Card. arXiv preprint arXiv:2410.21276.

[4] Levine, D.M., Tuwani, R., Kompa, B., Varma, A., Finlayson, S.G., Mehrotra, A. and Beam, A., 2023. The diagnostic and triage accuracy of the GPT-3 artificial intelligence model. MedRxiv.

[5] McCartney A., 2024. When Not to Use Generative AI. Gartner, Inc.

[6] Phan L., Khoja A., Mazeika M., Hendrycks D., 2024. LLMs Are Superhuman Forecasters. Pre-print.

[7] Radford, A., Narasimhan, K., Salimans, T. and Sutskever, I., 2018. Improving language understanding by generative pre-training.

[8] Schoenegger, P., Tuminauskaite, I., Park, P.S. and Tetlock, P.E., 2024. Wisdom of the silicon crowd: LLM ensemble prediction capabilities match human crowd accuracy. arXiv preprint arXiv:2402.19379.

[9] Soru, T., 2023, Spotting Signals in Text via Natural Language Understanding, Knowledge Graph Conference (KGC), New York City. doi: 10.5281/zenodo.7908658.

[10] Soru, T. and Marshall, J., 2024, February. Trend Extraction and Analysis via Large Language Models. In 2024 IEEE 18th International Conference on Semantic Computing (ICSC) (pp. 285-288). IEEE.